

A RIGOROUS CORRECTION AND A LOW-LOAD CONVERGENCE THEOREM FOR THE $SQ(d)$ HYDRODYNAMIC EQUATIONS WITH GENERAL SERVICE TIMES

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ABSTRACT. Status: partial progress, not a solution of the full open problem. The hydrodynamic equations of Aghajani and Ramanan describe the mean-field dynamics of the $SQ(d)$ load-balancing model with general service times. Agarwal and Ramanan asked whether solutions converge to the physical invariant state for a large class of initial conditions. We prove such convergence when the arrival rate is sufficiently small, the service distribution has an exponential moment, and the initial tagged-queue law has a finite exponential unfinished-work moment. The conclusion holds for every fixed $d \geq 2$, requires no monotonicity of the hazard rate, and gives convergence in a weighted total-variation norm, hence total-variation convergence of every age measure and convergence of the mean queue length.

The key correction is to discard two invalid ingredients of the earlier draft: the survivor-test “norm,” which is degenerate even for exponential service, and an exponential moment of the mean residual life, which can grow without bound for heavy-tailed service. For a tagged queue with n jobs and service age a , we instead use the exact conditional transform

$$V_\theta(n, a) = m_\theta^{n-1} \mathbb{E}[e^{\theta(S-a)} \mid S > a], \quad m_\theta = \mathbb{E}e^{\theta S}.$$

Its service drift is exactly $-\theta V_\theta$. A uniform Harris contraction for every prescribed environment path, combined with an explicit Lipschitz estimate for the nonlinear routing feedback, yields a genuine small-gain contraction at low load.

1. SCOPE AND STATUS OF THE RESULT

Agarwal and Ramanan characterize physical invariant states of the measure-valued hydrodynamic equations for the $SQ(d)$ model and identify convergence of the hydrodynamic dynamics to such an invariant state as a sufficient ingredient for the steady-state limit problem; see [1, Section 1.1]. Their question concerns a broad range of loads $\lambda < 1$, service laws, and initial conditions.

The theorem proved here is a genuine but restricted result. It establishes global convergence on an exponential-moment class for all sufficiently small $\lambda > 0$. It does *not* establish convergence for every $\lambda < 1$, and it does not cover service distributions without an exponential moment. Thus it should be classified as partial progress rather than a solution of the authors’ full problem.

The proof also explains precisely how to repair the earlier attempted argument. The state is first rewritten as the law of an age-dependent tagged queue. The correct Lyapunov function is not an exponential of the mean residual life. It is the conditional moment generating function of the *entire unfinished workload* of the tagged queue. This function has an exact negative service drift and provides both a queue-length tail bound and a uniform regeneration estimate. A weighted total-variation metric then permits a self-contained Harris argument. Finally, the polynomial routing rates are Lipschitz in the queue-length tails, and a low-load small-gain inequality closes the nonlinear feedback.

2. HYDRODYNAMIC EQUATIONS AND THE TAGGED-QUEUE LAW

Fix $d \geq 2$ and $\lambda > 0$. Let G be an absolutely continuous distribution on $[0, \infty)$ with density g , survivor function $\bar{G} = 1 - G$, right endpoint

$$L = \sup\{x \geq 0 : \bar{G}(x) > 0\} \in (0, \infty],$$

and mean one. We assume $G(0) = 0$. Its hazard rate is

$$h(a) = \frac{g(a)}{\bar{G}(a)}, \quad 0 \leq a < L.$$

We impose the standard well-posedness assumption used for the hydrodynamic equations.

Assumption 2.1 (Well-posed hydrodynamics). *For every initial state in the hydrodynamic state space, the hydrodynamic equations have a unique solution. In particular, this holds under the sufficient regularity hypotheses in [2]; local boundedness of g is the principal condition used in the uniqueness argument.*

For $0 \leq y \leq x \leq 1$, let

$$P_d(x, y) = \frac{x^d - y^d}{x - y} = \sum_{j=0}^{d-1} x^j y^{d-1-j},$$

with its polynomial value on the diagonal. A hydrodynamic state is an ordered sequence $\nu = (\nu_\ell)_{\ell \geq 1}$ of subprobability measures on $[0, L)$. Put

$$s_\ell = \langle 1, \nu_\ell \rangle, \quad s_0 = 1.$$

The state is called *proper* when $s_\ell \downarrow 0$. For a proper state, define its exact-length law by

$$(1) \quad \mu_0 = 1 - s_1, \quad \mu_n = \nu_n - \nu_{n+1}, \quad n \geq 1.$$

Here μ_0 is the mass of an idle queue, while μ_n is the age measure of queues containing exactly n jobs. Properness gives

$$\mu_0 + \sum_{n \geq 1} \mu_n([0, L)) = 1.$$

Thus μ is a probability law on

$$\mathbf{E} = \{\partial\} \cup (\mathbb{N} \times [0, L)),$$

where ∂ is the idle state. Conversely,

$$(2) \quad \nu_\ell(B) = \sum_{n \geq \ell} \mu_n(B), \quad s_\ell = \mu\{(n, a) : n \geq \ell\}.$$

Given a tail vector $s = (s_n)_{n \geq 0}$, define the tagged-queue arrival rates

$$(3) \quad \alpha_n(s) = \lambda P_d(s_n, s_{n+1}), \quad n \geq 0.$$

Since $0 \leq s_{n+1} \leq s_n \leq 1$,

$$(4) \quad 0 \leq \alpha_n(s) \leq \lambda d.$$

Moreover, if $p_n = s_n - s_{n+1}$, then

$$(5) \quad p_n \alpha_n(s) = \lambda(s_n^d - s_{n+1}^d).$$

The following exact-length formulation is the useful Markovian representation of the hydrodynamic equations. It is a direct reformulation of the equations in [2, 1].

Proposition 2.2 (Tagged nonlinear queue). *Let $\nu(t)$ be a proper hydrodynamic solution and let $\mu(t)$ be defined by (1). For every bounded test function φ having bounded continuous age derivative and finite support in the queue-length coordinate,*

$$(6) \quad \begin{aligned} \frac{d}{dt} \langle \varphi, \mu(t) \rangle &= \mu_0(t) \alpha_0(s(t)) (\varphi(1, 0) - \varphi(\partial)) \\ &\quad + \sum_{n \geq 1} \int_{[0, L)} \left[\partial_a \varphi(n, a) + \alpha_n(s(t)) (\varphi(n+1, a) - \varphi(n, a)) \right. \\ &\quad \left. + h(a) (\varphi(c(n)) - \varphi(n, a)) \right] \mu_n(t) (da), \end{aligned}$$

where $c(1) = \partial$ and $c(n) = (n-1, 0)$ for $n \geq 2$. Conversely, a probability-law solution of (6), with s recovered from (2), induces a proper solution of the hydrodynamic equations.

Proof. Subtract the level- $(n+1)$ age equation from the level- n age equation. An arrival to a queue of exact length n transfers its exact-length mass from n to $n+1$ without changing the service age, and its per-queue rate is (3) by (5). A service completion from exact length n sends the queue to ∂ when $n=1$ and to $(n-1, 0)$ when $n \geq 2$. The age of the job in service increases at unit speed between jumps. These three mechanisms give (6). Conversely, summing the exact-length equations over $n \geq \ell$ recovers the level- ℓ mass equation, and the method of characteristics for the age transport and killing term recovers the measure-valued age equation. Uniqueness in Assumption 2.1 identifies the two formulations. $\square$

For a deterministic environment path $s(\cdot)$, equation (6) is the forward equation of a time-inhomogeneous Markov process. Its transition kernel from time u to time t is denoted by $P_{u,t}^s$. The process is nonexplosive because its arrival rate is bounded by λd and a service completion can occur only for an initially present job or a job that has arrived.

3. THE CORRECT EXPONENTIAL UNFINISHED-WORK LYAPUNOV FUNCTION

Fix $\theta > 0$ such that

$$(7) \quad m = m_\theta := \mathbb{E} e^{\theta S} < \infty,$$

where S has distribution G . Since $\mathbb{E} S = 1$, one has $m > 1$. For $0 \leq a < L$, define

$$(8) \quad \chi(a) = \chi_\theta(a) := \frac{1}{\bar{G}(a)} \int_a^L e^{\theta(x-a)} g(x) dx = \mathbb{E}[e^{\theta(S-a)} \mid S > a].$$

Set

$$(9) \quad V(\partial) = 1, \quad V(n, a) = m^{n-1} \chi(a), \quad n \geq 1.$$

The interpretation is exact: conditional on state (n, a) , the unfinished work is the residual service of the job in service plus $n-1$ independent fresh service requirements. Therefore $V(n, a)$ is the conditional moment generating function of that total unfinished work. In particular,

$$(10) \quad V(n, a) \geq m^{n-1}, \quad n \geq 1.$$

The next identity is the central correction to the earlier draft.

Lemma 3.1 (Exact service-drift identity). *The function χ is locally absolutely continuous and, for almost every $a < L$,*

$$(11) \quad \chi'(a) + h(a)(1 - \chi(a)) = -\theta \chi(a).$$

Consequently, the age drift and service-completion terms in the tagged generator applied to V sum exactly to $-\theta V(n, a)$ for every busy state (n, a) .

Proof. Let

$$J(a) = \int_a^L e^{\theta(x-a)} g(x) dx.$$

Then $J'(a) = -g(a) - \theta J(a)$ almost everywhere. Since $\overline{G}'(a) = -g(a)$ and $h = g/\overline{G}$,

$$\chi'(a) = \frac{J'(a)}{\overline{G}(a)} - \frac{J(a)\overline{G}'(a)}{\overline{G}(a)^2} = -\theta\chi(a) + h(a)(\chi(a) - 1),$$

which is (11). Also $\chi(0) = m$. For $n \geq 2$,

$$V(n-1, 0) = m^{n-2}\chi(0) = m^{n-1},$$

and for $n = 1$, $V(\partial) = 1 = m^{n-1}$. Thus in both cases

$$\begin{aligned} \partial_a V(n, a) + h(a)(V(c(n)) - V(n, a)) &= m^{n-1}[\chi'(a) + h(a)(1 - \chi(a))] \\ &= -\theta V(n, a). \end{aligned}$$

□

Proposition 3.2 (Uniform exponential-moment drift). *Let $s(\cdot)$ be any deterministic environment path. Put*

$$(12) \quad \beta = \lambda d, \quad c = \theta - \beta(m-1).$$

If $c > 0$, then for all $0 \leq u \leq t$ and every state $x \in \mathbf{E}$,

$$(13) \quad P_{u,t}^s V(x) \leq e^{-c(t-u)} V(x) + H(1 - e^{-c(t-u)}), \quad H = \frac{\theta}{c}.$$

The same inequality holds after integration against any initial law.

Proof. At a busy state, Lemma 3.1 and $V(n+1, a) = mV(n, a)$ give

$$\mathcal{L}_t^s V(n, a) = -\theta V(n, a) + \alpha_n(s(t))(m-1)V(n, a) \leq -cV(n, a).$$

At the idle state,

$$\mathcal{L}_t^s V(\partial) = \alpha_0(s(t))(m-1) \leq \beta(m-1) = \theta - c = -cV(\partial) + \theta.$$

Hence, on all of $\mathbf{E}$,

$$\mathcal{L}_t^s V \leq -cV + \theta.$$

Dynkin's formula, first for stopped processes and bounded truncations of V , followed by localization and monotone convergence, yields (13) by Gronwall's inequality. □

For a proper hydrodynamic state, define its exponential unfinished-work moment by

$$(14) \quad \mathcal{V}_\theta(\nu) := 1 - s_1 + \sum_{n \geq 1} m^{n-1} \int_{[0,L)} \chi(a) (\nu_n - \nu_{n+1})(da).$$

This is exactly $\mu(V)$. It is substantially stronger than a finite mean queue length, but it is a natural and verifiable exponential-workload condition. By (10), it implies an exponential queue-length tail.

4. UNIFORM CONTRACTION FOR A PRESCRIBED ENVIRONMENT

We now establish a Harris contraction uniformly over all deterministic environment paths. The proof is included because the nonlinear argument needs constants that are genuinely uniform.

For $\eta > 0$, put

$$w(x) = 1 + \eta V(x)$$

and define the weighted total-variation distance

$$(15) \quad D_\eta(\mu, \tilde{\mu}) := \int_{\mathbf{E}} w(x) |\mu - \tilde{\mu}|(dx).$$

Equivalently, this is the optimal transport cost for

$$(16) \quad d_\eta(x, y) = \begin{cases} 0, & x = y, \\ w(x) + w(y), & x \neq y. \end{cases}$$

Indeed, an optimal coupling matches the common part of the two measures on the diagonal and transports only the mutually singular remainders.

We choose constants once and for all. Let

$$(17) \quad \bar{\lambda} = \min \left\{ 1, \frac{\theta}{2d(m-1)} \right\}.$$

For $0 < \lambda \leq \bar{\lambda}$, Proposition 3.2 has

$$(18) \quad c \geq \frac{\theta}{2}, \quad H \leq 2.$$

Set

$$(19) \quad R = 8, \quad B = \frac{\log(2R)}{\theta}, \quad T = B + 1,$$

$$(20) \quad \varepsilon_0 = \frac{1}{2}e^{-d\bar{\lambda}T}, \quad \gamma_0 = e^{-\theta T/2}, \quad \eta = \min \left\{ 1, \frac{\varepsilon_0}{R+4} \right\},$$

and

$$(21) \quad \rho = \max \left\{ 1 - \frac{\varepsilon_0}{2}, 1 - \frac{\eta(1 - \gamma_0)(R - 4)}{2 + \eta R} \right\} < 1.$$

Lemma 4.1 (Uniform idle-state minorization). *For every $0 < \lambda \leq \bar{\lambda}$, every deterministic environment path $s(\cdot)$, and every $x \in \mathbf{E}$ with $V(x) \leq R$,*

$$(22) \quad P_{0,T}^s(x, \cdot) \geq \varepsilon_0 \delta_\partial(\cdot).$$

Proof. For $x = (n, a)$, let Y_x be the total unfinished work: the residual service time of the current job, conditional on age a , plus $n - 1$ independent service times with law G . By construction, $\mathbb{E}e^{\theta Y_x} = V(x)$. Hence, if $V(x) \leq R$, Markov's inequality and the definition of B give

$$\mathbb{P}(Y_x > B) \leq Re^{-\theta B} = \frac{1}{2}.$$

Use a rate- λd dominating Poisson clock for potential arrivals. If this clock has no point during $[0, T]$ and $Y_x \leq B < T$, the queue is idle at time T . The first event has probability at least $e^{-\lambda d T} \geq e^{-d\bar{\lambda} T}$ and is independent of the service requirements. Thus the probability of being idle at time T is at least $\frac{1}{2}e^{-d\bar{\lambda} T} = \varepsilon_0$. The idle initial state satisfies the same bound directly. $\square$

Proposition 4.2 (Uniform same-environment contraction). *For every $0 < \lambda \leq \bar{\lambda}$, every deterministic environment path $s(\cdot)$, and all probability laws $\mu, \tilde{\mu}$ with finite V -moment,*

$$(23) \quad D_\eta(\mu P_{0,T}^s, \tilde{\mu} P_{0,T}^s) \leq \rho D_\eta(\mu, \tilde{\mu}).$$

Proof. Write $P = P_{0,T}^s$. Proposition 3.2 gives

$$(24) \quad PV \leq \gamma_\lambda V + H_\lambda(1 - \gamma_\lambda), \quad \gamma_\lambda = e^{-cT} \leq \gamma_0, \quad H_\lambda = \frac{\theta}{c} \leq 2.$$

It is enough to construct, for every x, y , a coupling of $P(x, \cdot)$ and $P(y, \cdot)$ whose expected d_η -cost is at most $\rho d_\eta(x, y)$.

If $x = y$, use the identical coupling. Suppose $x \neq y$ and write $S = V(x) + V(y)$. First assume $S \leq R$. Lemma 4.1 permits a split coupling that sends both chains to ∂ with common probability ε_0 . On the complement, use any coupling. Its expected cost is at most

$$\begin{aligned} 2(1 - \varepsilon_0) + \eta(PV(x) + PV(y)) &\leq 2(1 - \varepsilon_0) + \eta(S + 4) \\ &\leq 2(1 - \varepsilon_0) + \eta(R + 4) \\ &\leq 2 - \varepsilon_0. \end{aligned}$$

Since $d_\eta(x, y) = 2 + \eta S \geq 2$, the contraction factor in this case is at most $1 - \varepsilon_0/2$.

Now assume $S > R$. Any coupling has expected cost at most

$$2 + \eta(PV(x) + PV(y)) \leq 2 + \eta[\gamma_\lambda S + 2H_\lambda(1 - \gamma_\lambda)].$$

Subtracting this expression from $2 + \eta S$ and using $H_\lambda \leq 2$ and $1 - \gamma_\lambda \geq 1 - \gamma_0$ gives a gap of at least

$$\eta(1 - \gamma_0)(S - 4).$$

The function $(S - 4)/(2 + \eta S)$ is increasing for $S > 4$. Hence the contraction factor is at most

$$1 - \frac{\eta(1 - \gamma_0)(R - 4)}{2 + \eta R}.$$

Taking the larger of the two factors gives (21). Integrating the pairwise couplings against an optimal initial coupling proves (23). $\square$

We also need two finite-time stability estimates. They are weaker than Proposition 4.2, but they quantify the effect of changing the environment.

Lemma 4.3 (Finite-time expansion and environment perturbation). *Assume $0 < \lambda \leq \bar{\lambda}$ and let $0 \leq t \leq T$.*

(i) *For every environment path $s(\cdot)$,*

$$(25) \quad D_\eta(\mu P_{0,t}^s, \tilde{\mu} P_{0,t}^s) \leq A D_\eta(\mu, \tilde{\mu}), \quad A = 1 + 2\eta.$$

(ii) *Let $s(\cdot)$ and $\tilde{s}(\cdot)$ be two environment paths and define*

$$(26) \quad \delta(u) = \sup_{n \geq 0} |\alpha_n(s(u)) - \alpha_n(\tilde{s}(u))|.$$

If $\mu(V) \leq 4$, then

$$(27) \quad D_\eta(\mu P_{0,t}^s, \mu P_{0,t}^{\tilde{s}}) \leq C \int_0^t \delta(u) du, \quad C = 2 + \eta(4(m + 1) + 4).$$

Proof. For (i), couple equal initial states identically. For unequal initial states x, y , use any coupling of the two evolutions. By Proposition 3.2, (18), and $t \leq T$,

$$P_{0,t}^s V(x) \leq V(x) + 2, \quad P_{0,t}^s V(y) \leq V(y) + 2.$$

Thus the expected cost is at most

$$2 + \eta(V(x) + V(y) + 4) \leq (1 + 2\eta)(2 + \eta(V(x) + V(y))).$$

Integrating over an optimal initial coupling gives (25).

For (ii), start the two processes from the same state and couple them identically until the first unmatched arrival. While they agree and have queue length n , the unmatched-arrival intensity is

$|\alpha_n(s(u)) - \alpha_n(\tilde{s}(u))| \leq \delta(u)$. At such an arrival, one state changes from x to the state obtained by appending one job. By (9), this multiplies V by exactly m . Thereafter, Proposition 3.2 bounds the sum of the two terminal V -moments by

$$(m+1)V(x) + 2H \leq (m+1)V(x) + 4.$$

Because the common process starts with V -moment at most 4, the drift bound keeps its V -moment at most 4 up to time T . Integrating the first-unmatched-arrival compensator therefore gives the cost bound

$$\int_0^t \delta(u) [2 + \eta(4(m+1) + 4)] \, du,$$

which is (27). This coupling argument also justifies the estimate for merely measurable environment paths by approximation. $\square$

5. CLOSING THE NONLINEAR FEEDBACK AT LOW LOAD

Let Φ_t denote the nonlinear hydrodynamic flow in exact-length form. Thus $\mu_t = \Phi_t(\mu_0)$ evolves with the environment generated by its own tails

$$s_n(t) = \mu_t\{(k, a) : k \geq n\}.$$

The routing map is uniformly Lipschitz. Indeed, on $[0, 1]^2$,

$$(28) \quad |P_d(x, y) - P_d(\tilde{x}, \tilde{y})| \leq \frac{d(d-1)}{2} (|x - \tilde{x}| + |y - \tilde{y}|),$$

because each partial derivative of the polynomial P_d is bounded by $d(d-1)/2$. Since the weighted total-variation distance dominates the difference of probabilities of every measurable event, two nonlinear solutions satisfy

$$(29) \quad \delta(t) \leq \lambda L_d D_\eta(\mu_t, \tilde{\mu}_t), \quad L_d = d(d-1).$$

Define

$$(30) \quad \lambda_0 = \min \left\{ \bar{\lambda}, \frac{1}{CL_d T}, \frac{1-\rho}{2eCAL_d T} \right\} > 0,$$

where A, C, T, ρ are given in (19)–(21) and Lemma 4.3. No optimization of this conservative constant is intended.

Proposition 5.1 (Nonlinear block contraction). *Suppose $0 < \lambda \leq \lambda_0$. If $\mu_0(V) \leq 4$ and $\tilde{\mu}_0(V) \leq 4$, then*

$$(31) \quad D_\eta(\Phi_T(\mu_0), \Phi_T(\tilde{\mu}_0)) \leq \kappa D_\eta(\mu_0, \tilde{\mu}_0), \quad \kappa = \frac{1+\rho}{2} < 1.$$

Proof. Let μ_t and $\tilde{\mu}_t$ be the two nonlinear solutions, with environment paths s and $\tilde{s}$, and write

$$D(t) = D_\eta(\mu_t, \tilde{\mu}_t).$$

The drift estimate and $H \leq 2$ imply that both V -moments remain at most 4 on $[0, T]$. For $t \leq T$, insert an intermediate evolution of $\tilde{\mu}_0$ in the environment s . Lemma 4.3, followed by (29), gives

$$(32) \quad D(t) \leq AD(0) + C\lambda L_d \int_0^t D(u) \, du.$$

Hence

$$(33) \quad D(t) \leq Ae^{C\lambda L_d t} D(0), \quad 0 \leq t \leq T.$$

At time T , use Proposition 4.2 for the same-environment part instead of the finite-time expansion. This yields

$$\begin{aligned} D(T) &\leq \rho D(0) + C\lambda L_d \int_0^T D(u) du \\ &\leq \left[\rho + C\lambda L_d AT e^{C\lambda L_d T} \right] D(0). \end{aligned}$$

The definition of λ_0 implies $C\lambda L_d T \leq 1$ and

$$C\lambda L_d AT e^{C\lambda L_d T} \leq e C\lambda L_d AT \leq \frac{1-\rho}{2}.$$

Thus the bracket is at most $(1+\rho)/2 = \kappa$. □

6. MAIN THEOREM

We can now state the corrected convergence result in hydrodynamic variables.

Theorem 6.1 (Low-load global convergence). *Assume Assumption 2.1, fix $d \geq 2$, and suppose that G has an exponential moment (7) for some $\theta > 0$. Let $\lambda_0 > 0$ be the constant in (30). For every $0 < \lambda \leq \lambda_0$, the following hold.*

(a) *There exists a unique physical invariant state ν^* of the $SQ(d)$ hydrodynamic equations. Its exact-length law μ^* satisfies*

$$(34) \quad \mu^*(V) \leq \frac{\theta}{\theta - \lambda d(m-1)} \leq 2.$$

(b) *Let $\nu(0)$ be any proper initial state satisfying*

$$(35) \quad \mathcal{V}_\theta(\nu(0)) < \infty,$$

and let $\nu(t)$ be its hydrodynamic solution. If $\mu(t)$ and μ^ are the associated exact-length laws, then*

$$(36) \quad \mathcal{D}_\eta(\mu(t), \mu^*) \longrightarrow 0 \quad \text{exponentially as } t \rightarrow \infty.$$

(c) *For every fixed $\ell \geq 1$,*

$$(37) \quad \|\nu_\ell(t) - \nu_\ell^*\|_{TV} \longrightarrow 0, \quad s_\ell(t) \longrightarrow s_\ell^*.$$

Moreover,

$$(38) \quad \sum_{\ell \geq 1} s_\ell(t) \longrightarrow \sum_{\ell \geq 1} s_\ell^*.$$

Proof. Let

$$\mathcal{P}_4 = \{\mu : \mu \text{ is a probability law on } \mathbb{E}, \mu(V) \leq 4\}.$$

The space of finite signed measures with finite w -variation is a Banach space, and $\mathcal{P}_4$ is a closed subset: total mass and positivity are closed, and $V/w \leq 1/\eta$ makes $\mu \mapsto \mu(V)$ continuous in $\mathcal{D}_\eta$. Thus $\mathcal{P}_4$ is complete. Proposition 3.2 and $H \leq 2$ show that Φ_T maps $\mathcal{P}_4$ into itself. Proposition 5.1 makes this map a strict contraction. Banach's fixed-point theorem therefore gives a unique $\mu^* \in \mathcal{P}_4$ satisfying $\Phi_T(\mu^*) = \mu^*$.

The nonlinear flow is autonomous. For every $r \geq 0$, $\Phi_r(\mu^*)$ also lies in $\mathcal{P}_4$ and

$$\Phi_T(\Phi_r(\mu^*)) = \Phi_r(\Phi_T(\mu^*)) = \Phi_r(\mu^*).$$

Uniqueness of the fixed point of Φ_T implies $\Phi_r(\mu^*) = \mu^*$ for every r . Its tail-age projection is therefore a physical invariant hydrodynamic state ν^* .

We next prove uniqueness among all physical invariant states, without assuming a moment bound. Let $\bar{\nu}$ be physical and invariant, and let $\bar{\mu}$ be its exact-length probability law. It is invariant for the

tagged nonlinear flow. Write $H = \theta/[\theta - \lambda d(m-1)]$. Iterating (13) over kT and using invariance, for every truncation level $M_0 < \infty$,

$$\begin{aligned}\bar{\mu}(V \wedge M_0) &= \bar{\mu}(P_{0,kT}(V \wedge M_0)) \\ &\leq \int \min \left\{ e^{-ckT} V(x) + H, M_0 \right\} \bar{\mu}(dx).\end{aligned}$$

For fixed M_0 , dominated convergence as $k \rightarrow \infty$ yields $\bar{\mu}(V \wedge M_0) \leq H \wedge M_0$. Letting $M_0 \rightarrow \infty$ gives $\bar{\mu}(V) \leq H \leq 2$. Hence $\bar{\mu} \in \mathcal{P}_4$ and is a fixed point of Φ_T , so $\bar{\mu} = \mu^*$. This proves (a), including (34).

Now let μ_0 satisfy (35). Proposition 3.2 gives

$$\mu_t(V) \leq e^{-ct} \mu_0(V) + H(1 - e^{-ct}).$$

Choose $t_0 < \infty$ so that $e^{-ct_0} \mu_0(V) \leq 2$. Then $\mu_t(V) \leq 4$ for every $t \geq t_0$. At the block times,

$$D_\eta(\mu_{t_0+kT}, \mu^*) \leq \kappa^k D_\eta(\mu_{t_0}, \mu^*).$$

For an intermediate time $t = t_0 + kT + r$, $0 \leq r < T$, the argument leading to (33) gives

$$D_\eta(\mu_t, \mu^*) \leq A e^{C\lambda L_d T} \kappa^k D_\eta(\mu_{t_0}, \mu^*).$$

This proves the exponential convergence in (b).

Finally, for every bounded measurable f with $|f| \leq 1$,

$$\langle f, \nu_\ell(t) - \nu_\ell^* \rangle = \int_{\mathbf{E}} f(a) \mathbf{1}_{\{n \geq \ell\}} (\mu_t - \mu^*)(d(n, a)).$$

Therefore

$$\|\nu_\ell(t) - \nu_\ell^*\|_{TV} \leq D_\eta(\mu_t, \mu^*),$$

which proves (37). In addition, (10) and the uniform bound $\mu_t(V) \leq 4$ for $t \geq t_0$ imply a uniform exponential moment of the queue length N_t :

$$\mathbb{E} [m^{N_t-1} \mathbf{1}_{\{N_t \geq 1\}}] \leq 4.$$

Hence $\{N_t : t \geq t_0\}$ is uniformly integrable. Weighted total-variation convergence implies convergence in distribution of N_t to N^* , so uniform integrability yields $\mathbb{E} N_t \rightarrow \mathbb{E} N^*$. Since $\mathbb{E} N_t = \sum_{\ell \geq 1} s_\ell(t)$, this is (38). $\square$

Remark 6.2 (What the theorem does and does not settle). Theorem 6.1 gives a complete proof in a nontrivial low-load regime and, in that regime, also proves uniqueness of the physical invariant state for every $d \geq 2$. The proof is robust to nonmonotone and unbounded hazard rates whenever the tagged process and hydrodynamic equations are well posed. It does not settle the principal high-load difficulty: as λ grows, the Lipschitz feedback term need not be smaller than the prescribed-environment Harris contraction. New structure, rather than a refinement of constants alone, appears necessary to reach all $\lambda < 1$.

7. ADDITIONAL EXACT BALANCE IDENTITY

The tagged representation also yields a useful workload balance that is independent of the small-load contraction. Let $W(n, a)$ denote the conditional mean unfinished work in state (n, a) :

$$W(\partial) = 0, \quad W(n, a) = n - 1 + R(a), \quad R(a) = \mathbb{E}[S - a \mid S > a].$$

Whenever the relevant first moments are finite, the mean unfinished workload $\mathcal{W}(t) = \mu_t(W)$ satisfies

$$(39) \quad \frac{d}{dt} \mathcal{W}(t) = \lambda - s_1(t).$$

Indeed, unfinished work decreases at rate one whenever the tagged queue is busy, while the average arrival rate to a tagged queue is

$$\sum_{n \geq 0} (s_n - s_{n+1}) \lambda P_d(s_n, s_{n+1}) = \lambda \sum_{n \geq 0} (s_n^d - s_{n+1}^d) = \lambda.$$

Identity (39) is not by itself a Lyapunov inequality, because $s_1(t)$ can be below λ . It nevertheless shows that every invariant proper state with finite mean unfinished work must satisfy $s_1^* = \lambda$, in agreement with [1].

8. REMAINING OPEN DIRECTIONS

The following extensions are not proved here.

- (i) Remove the low-load condition and prove convergence for every $\lambda < 1$.
- (ii) Replace the exponential service and initial-work moments by substantially weaker moment or tightness assumptions.
- (iii) Establish a contraction or order structure that uses the average arrival rate λ , rather than the worst-case tagged rate λd .
- (iv) Combine hydrodynamic convergence with uniform stationary tightness to obtain the full interchange-of-limits statement for the stationary N -server systems.

The first item is the central unresolved part of the Agarwal–Ramanan problem. The weighted unfinished-work transform in this paper supplies a correct compactness and regeneration mechanism; the remaining obstruction is control of the nonlinear feedback outside the perturbative low-load regime.

APPENDIX A. WHY THE DISCARDED PROOF CANNOT BE REPAIRED LOCALLY

For completeness, we record two explicit counterexamples to foundational claims in the earlier draft. They motivate the replacement developed above.

Proposition A.1 (The survivor-test functional is not a norm). *Define*

$$\vartheta_r(a) = \frac{\overline{G}(a+r)}{\overline{G}(a)}, \quad \|\zeta\|_{\mathcal{R}} = \sup_{r \geq 0} |\langle \vartheta_r, \zeta \rangle|.$$

This functional need not separate finite signed measures, even when G is absolutely continuous and nonlattice.

Proof. For unit-mean exponential service, $\overline{G}(a) = e^{-a}$ and $\vartheta_r(a) = e^{-r}$ is independent of a . Hence

$$\|\zeta\|_{\mathcal{R}} = |\zeta([0, \infty))|.$$

In particular, $\|\delta_0 - \delta_1\|_{\mathcal{R}} = 0$ although $\delta_0 \neq \delta_1$. Thus the functional is only a seminorm and cannot imply weak convergence of age measures. $\square$

Proposition A.2 (An exponential mean-residual-life bound is false). *There are unit-mean absolutely continuous service distributions and admissible one-job initial states for which*

$$\sup_{t \geq 0} \int e^{\theta R(a)} \nu_1(t)(da) = \infty$$

for every $\theta > 0$, where $R(a) = \mathbb{E}[S - a \mid S > a]$.

Proof. Fix $\alpha > 1$, put $b = \alpha - 1$, and take the unit-mean Lomax distribution

$$\overline{G}(a) = \left(1 + \frac{a}{b}\right)^{-\alpha}.$$

A direct calculation gives

$$R(a) = 1 + \frac{a}{b}.$$

Start with one age-zero job at every queue and no waiting jobs, so $\nu_1(0) = \delta_0$ and $\nu_\ell(0) = 0$ for $\ell \geq 2$. The initial cohort contributes the measure $\overline{G}(t)\delta_t$ to $\nu_1(t)$. Therefore

$$\int e^{\theta R(a)} \nu_1(t)(da) \geq e^{\theta(1+t/b)} \left(1 + \frac{t}{b}\right)^{-\alpha} \longrightarrow \infty.$$

Thus no uniform exponential bound of the claimed form can follow from finite mean service and an initial exponential envelope. The transform V_θ in (9) avoids this error because it uses the true conditional exponential residual-work moment, and it is assumed finite only when the service law itself has an exponential moment. $\square$

Remark A.3 (The additive-tail absorption error). A bound of the form

$$D(T) \leq (1 - \varepsilon)D(0) + C\delta$$

cannot be converted into a strict homogeneous contraction by choosing a fixed small δ , because $D(0)$ may be arbitrarily close to zero. Proposition 4.2 avoids this issue by working in a weighted total-variation distance for which the drift and minorization combine into a truly homogeneous contraction.

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